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YASH PATEL,[1] PRAJAKTA BELEKAR,[1] GEORG LADURNER,[1] SYBREN WORM,[1] LUCAS MAY,[1] MARIA VARAKA,[1] THERESA ROHRMOSER,[1] FERNANDO GARCÍA-RAMÍREZ,[1] RENÉ WERKMEISTER,[1] BERNHARD BAUMANN,[1,2] AND CONRAD W. MERKLE[1,*]

[1] *Center for Medical Physics and Biomedical Engineering, Medical University of Vienna, Währinger Gürtel 18-20/4L, 1090 Vienna, Austria*

[2] *Institute of Biomedical Physics, Medical University of Innsbruck, Müllerstraße 44, 6020 Innsbruck, Austria*

[*] *conrad.merkle@meduniwien.ac.at*

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1. Introduction

Optical coherence tomography angiography (OCTA) is an extension of OCT that visualizes vascular perfusion using the motion contrast from red blood cells (RBCs). Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [1]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones, or it can be used to examine individual vessels with measurements of diameter or tortuosity. Nevertheless, OCT is prone to certain artifacts which specifically affect visualization of the vasculature. These artifacts may lead to underestimations of measurements of vessel diameter and density. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the center than at the sides (Figure 1A). This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels. To compensate for these artifacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished. Here we examine different methods of measuring vessel size from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. We finally propose ways of mitigating or correcting for underestimations of vessel diameter related to typical OCT artifacts.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample. OCT has found widespread clinical use in ophthalmology with XXX number of systems and XXX OCT scans performed annually around the world. OCT angiography (OCTA) is a functional extension of OCT which highlights motion between repeated OCT scans to generate images of vascular networks. Typically, OCTA signals are first calculated from the OCT intensity and/or phase information, segmented in 3D manually, algorithmically, or using a deep learning approach, and finally compressed into a 2D image using a maximum intensity projection. These 2D vascular maps can then be binarised by using a global or adaptive threshold, and finally vascular metrics

such as vessel density and diameter can be computed. Of these processing stages, the thresholding step is perhaps the most important. There are dozens of different thresholding methods that have been proposed over the years and each of these may produce different results in the binary map, which would in turn influence the quantification of vascular metrics. In this work, we examine an alternate approach for extracting quantitative vessel diameter by fitting a model to the OCTA data before binarisation to avoid thresholding-based effects.

In addition to thresholding-based variability, we hypothesize that OCTA metrics are also affected by OCT artifacts caused by the non-uniform backscattering profiles of red blood cells (RBCs). Because RBCs have a distinct bi-concave disc shape, they more effectively backscatter light when aligned perpendicular to the beam, rather than parallel to it. It has already been observed that in large vessels, a so-called hourglass artifact appears when viewed cross-sectionally. This is caused by the alignment of the RBCs with the vessel walls while under laminar flow, where the RBCs are perpendicular to the beam in the center of the vessels and parallel to the beam at the edges of the vessels. This artifact artificially suppresses the OCT and OCTA signals at the edges of the vessels, making them appear more narrow when projected into an en-face view, which is the preferred way of displaying angiographic data. Here we use contrast enhanced OCT (CE-OCT) to investigate how significant these effects are and how they can be corrected.

2. Methods and Materials

2.1. Imaging system

All imaging was performed using a custom-built polarisation-sensitive spectral-domain OCT ophthalmoscope, described in detail previously [2]. The system was centred at 840 nm with a spectral bandwidth of 100 nm at full width half maximum (FWHM), providing an axial resolution of approximately 5.1 µm in air and 3.8 µm in tissue (assuming a refractive index of 1.35). A sensitivity of 96 dB was achieved at A-scan rate of 80 kHz and a beam diameter of 0.5 mm at the pupil plane (power at cornea = 2.85 mW) [3]. This system was employed to acquire volumetric OCT data from mouse retinas before, during, and after an intravascular contrast agent administration, as well as before and after functional stimulation of the mouse retina using a 502 nm light-emitting diode (LED) ring flashing at 10 Hz (power at cornea = 26.6 μW). This wavelength was selected to target the peak absorption of mouse rhodopsin (~500 nm), enabling predominantly rod-driven photoreceptor stimulation [4, 5].

2.2. Experimental protocol

Three mouse models were used in this study: VLDLR knockout mice (B6;129S7-*Vldlr*^{tm1Her}/J, The Jackson Laboratory, Bar Harbor, USA), C57BL/6 mice (C57BL/6, The Jackson Laboratory, Bar Harbor, USA), and SOD1 transgenic mice (strain details, The Jackson Laboratory, Bar Harbor, USA), with n = XX mice per group. All imaging sessions were conducted under general anaesthesia using either isoflurane or a ketamine/xylazine combination. Isoflurane was induced at 4% in oxygen for 4 minutes and subsequently maintained at 2% in oxygen at a flow rate of 1.5–2 L/min. In cases where isoflurane alone did not provide adequate sedation, an intraperitoneal injection of ketamine/xylazine cocktail (100 mg/kg ketamine, 6 mg/kg xylazine; 10 mL/kg body weight) was administered. To maintain core body temperature of the mice throughout imaging, they were positioned on a heating pad. Mice pupils were dilated with one drop of tropicamide (5 mg/mL, Agepha Pharma s.r.o., Senec, Slovakia) per eye, and artificial tear drops (Oculotect, ALCON Pharma GmbH, Freiburg im Breisgau, Germany) were applied at regular intervals to prevent cornea from drying out.

Intralipid has been shown to be a highly scattering contrast agent that fills the vascular lumen uniformly, thereby compensating for the artefact caused by the diminished backscattering at vessel peripheries caused by RBC orientation effects [3, 6, 7]. For this study, Intralipid 20%

(3 mL/kg body weight) was administered as a bolus injection via the mice tail vein. Volumetric angiography data were acquired over a 1 mm × 1 mm field of view centred on the optic nerve head (ONH), with a scan density of 500 A-scans per B-scan and five repeated B-scans at each of 400 slow-axis positions (acquisition time approximately 16 seconds). For this study, pre- and post-injection datasets and pre- and post-stimulation datasets were acquired. Additionally, a dynamic-contrast OCT (DyC-OCT) protocol [8] was employed pre- and post-contrast injection, consisting of 2000 repeated B-scans at the same cross-sectional location (interscan time 8.2 ms) over a 1 mm lateral field of view. All animal procedures were approved by the ethics committee of the Medical University of Vienna and the Austrian Federal Ministry of Education, Science, and Research98 (BMBWF/66.009/0272-V/3b/2019).

2.3. Post-processing

Cross-polarised OCT intensity data from the retinal pigment epithelium (RPE) was used for inter-frame alignment and B-scan flattening [3, 9]. Angiograms were derived by applying the complex subtraction OCT angiogram (OCTA) algorithm [10, 11]. To enhance vessel visibility and suppress background noise, a 3D averaging filter with a kernel size of 50 × 50 × 10 voxels was applied to both the angiogram and DyC-OCT volumes. For the volumetric angiogram data, the kernel dimensions correspond to the lateral fast-axis (x), lateral slow-axis (y), and axial (z) dimensions, while for the DyC-OCT data, the kernel spans the lateral (x), axial (z), and temporal (t) dimensions. All data processing and analysis were performed in MATLAB (R2019b, MathWorks, Natick, MA, USA).

2.4. Diameter measurement

All vessel diameter measurements were performed on flattened B-scan angiograms. In this study, we compare three projection-based approaches and two model-based approaches for measuring vessel diameter from OCTA data, using post-contrast measurements as a ground truth reference. All measurements were performed in linear intensity scale. Vessel intensity profiles were measured from en-face intensity profiles generated by projecting the A-scans within a rectangular region of interest (ROI) encompassing the vessel cross-section along the lateral (x) and axial (z) directions. Vessel diameter was then calculated by fitting parametric models to these one-dimensional intensity profiles, extracted from the B-scan data across each vessel of interest. The details about the same are described in the following sections.

2.4.1. Ground truth calculation

Ground truth vessel diameters were obtained from contrast-enhanced B-scan angiograms, where the uniformly scattering Intralipid fills the full vascular lumen regardless of RBC orientation. For each vessel of interest, an elliptical ROI was manually selected around the vessel cross-section in the flattened B-scan. The lateral axis of the ellipse (perpendicular to the OCT beam) was taken as the ground truth vessel diameter, as this corresponds to the dimension represented in en-face OCTA projection maps. In cases where the lateral axis length exceeded the axial axis length (i.e. when the lateral dimension was the major axis of the ellipse), eccentricity correction was performed by correcting the measured lateral length by multiplying with the ratio of the minor to the major axis length. This correction compensated for the apparent broadening of the vessel cross-section caused by an oblique intersection of the imaging plane with the vessel axis.

2.4.2. Vessel projection calculation

For each vessel of interest, a rectangular ROI encompassing the vessel cross-section was selected from the DyC-OCT B-scan angiograms. En-face intensity profiles were then generated by projecting the A-scans within this ROI along the lateral (x) and axial (z) directions. Two projection methods were compared: maximum intensity projection (MaIP), which selects the

highest intensity value along each A-scan, and mean intensity projection (MeIP), which averages the values along each A-scan. The resulting one-dimensional profiles were normalised to a range of 0–1 to ensure uniformity across vessels, scans, and animals. Additionally, for comparison with conventional OCTA processing pipelines that rely on image binarisation, a binary mask was generated by applying a global threshold to the en-face projection maps. These three approaches: mean projection, maximum projection, and binary thresholding, represent commonly used methods in the literature for OCTA quantification [12–15].

2.4.3. Vessel diameter calculation

Two parametric models were then fitted to these one-dimensional intensity profiles from the previous step, using nonlinear least-squares regression in MATLAB.

The first model was a standard Gaussian function:

$$f(x) = A \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) + C \tag{1}$$

where A is the peak amplitude, μ is the centre position, σ is the standard deviation, and C is a constant baseline offset.

The second model was a generalised Gaussian (GG) function:

$$f(x) = A \exp\left(-\left(\frac{|x-\mu|}{\alpha}\right)^{\beta}\right) + C \tag{2}$$

where A is the peak amplitude, μ is the centre position, α is the scale parameter, β is the shape parameter, and C is a constant baseline offset. The shape parameter β was constrained to $\beta \geq 2$ during fitting. Given the imaging system configuration, all vessels are expected to produce at minimum a Gaussian intensity profile in the B-scan cross-section, and so profiles sharper than Gaussian profile ($\beta < 2$) are therefore not technically meaningful. With this constraint, the GG accommodates profiles ranging from Gaussian ($\beta = 2$, where $\sigma = \alpha/\sqrt{2}$) to increasingly flat-topped profiles ($\beta > 2$).

Two standard width metrics were calculated from the fitted models: the full width at half maximum (FWHM) and the $1/e^2$ width. For the standard Gaussian fit, the $1/e^2$ width was calculated as:

$$w_{1/e^2} = 4\sigma \tag{3}$$

The FWHM for a Gaussian is related to the $1/e^2$ width by a fixed proportionality:

$$\mathrm{FWHM} = 2\sigma\sqrt{2\ln 2} = \frac{\sqrt{2\ln 2}}{2}\, w_{1/e^2} \tag{4}$$

and therefore only the $1/e^2$ width was calculated directly from the fit, with the FWHM obtainable via this known relationship.

For the GG fit, no such linear relationship exists between the FWHM and $1/e^2$ width, since both depend on the shape parameter β. Both metrics were therefore calculated independently as:

$$\mathrm{FWHM}_{\mathrm{GG}} = 2\alpha(\ln 2)^{1/\beta} \tag{5}$$

$$w_{1/e^2,\,\mathrm{GG}} = 2\alpha \cdot 2^{1/\beta} \tag{6}$$

For the binary mask approach, multiple threshold values were used, and the resulting vessel diameters were compared to the ground truth. For each C-scan, the maximum angiogram intensity within the smoothed volume was calculated, and the threshold values were a fixed percentage of this maximum intensity. The vessel diameter was then measured as the number of pixels above each of the threshold values across the vessel cross-section.

3. Results

3.1. Repeatability and Linearity

-For BW methods, max and mean projections show opposite trends repeatability and linearity as a function of threshold. -Mean intensity projection typically shows improved measurements

4. Discussion

4.1. Maximum Intensity or Mean Intensity?

4.2. Future Directions

A future direction of this work to to move toward fully automated segmentation and calculation vessel diameters from enface angiogram maps. While ROIs were manually selected in this work, it is also possible to automatically segment angiographic data using traditional skeletonization methods, which are commonly used for automated OCTA analysis. From a skeletonized image, vessel angle can be computed for further extraction of vascular cross-sections perpendicular to the vessel axis. These cross-sections could then be processed using the model-based methods described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-automated approach shown here for 2D OCTA data. This type of model-based approach is particularly important for comparing 3D volumes acquired at different time points in the mouse eye as there are often intensity losses caused by drying of the eye, cataract formation, etc. As we have shown in this work, even low levels of intensity loss occurring over a short time window can significantly impact the measured vessel diameter when using a binarization approach, but not when using the models presented here.

4.3. Computation Time

It is clear from the results that model-based approaches outperform traditional diameter measurements based on a binarized image. Nevertheless, the improved linearity and repeatability come at the cost of computation time. Fitting a model to the raw aniogram data is significantly more computationally expensive and may not always be the most appropriate choice if results of the processing are time sensitive. Nevertheless, these approaches are highly parallelizable given the independent nature of the measurements, so it is expected that compuation time could be significantly accelerated with appropriate hardware.

4.4. Threshholding

The results presented here used a basic, global threshold for the binarization of the angiogram data. There are numerous other approaches for binarizing the angiogram, which will most likely each behave differently. For example, angiogram signals can be enhanced with vesselness filters and dynamic thresholds can be used to binarize images with varying SNR. When applying a vesselness filter, it is possible to enhance the shape of the angiogram and omit noise, but it is also possible for such enhancement to overcompensate the signal, leading to larger than expected vessel appearance. Dynamic thresholding can also yield a more uniform angiogram appearance, but similarly risks over- or undercompensating vessels given that threshold level directly affects the measured diameter. Further analysis of these other methods would be required to determine their efficacy in returning accurate and precise diameter values.

References

1. A. Javed, A. Khanna, E. Palmer, *et al.*, "Optical coherence tomography angiography: A review of the current literature," J. Int. Med. Res. **51**, 03000605231187933 (2023). Doi: 10.1177/03000605231187933.
2. S. Fialová, M. Augustin, M. Glösmann, *et al.*, "Polarization properties of single layers in the posterior eyes of mice and rats investigated using high resolution polarization sensitive optical coherence tomography," Biomed. Opt. Express **7**, 1479–1495 (2016).

3. C. W. Merkle, M. Augustin, D. J. Harper, *et al.*, "High-resolution, depth-resolved vascular leakage measurements using contrast-enhanced, correlation-gated optical coherence tomography in mice," Biomed. Opt. Express **12**, 1774–1791 (2021).
4. T. Son, B. Wang, Y. Lu, *et al.*, "Concurrent OCT imaging of stimulus evoked retinal neural activation and hemodynamic responses," Proc. SPIE–the Int. Soc. for Opt. Eng. **10045**, 1004522 (2017).
5. T. Son, G. Ma, and X. Yao, "Functional OCT reveals anisotropic changes of retinal flicker-evoked vasodilation," Opt. Lett. **49**, 2121–2124 (2024).
6. Y. Pan, J. You, N. D. Volkow, *et al.*, "Ultrasensitive detection of 3D cerebral microvascular network dynamics in vivo," NeuroImage **103**, 492–501 (2014).
7. P. Cimalla, J. Walther, M. Mittasch, and E. Koch, "Shear flow-induced optical inhomogeneity of blood assessed in vivo and in vitro by spectral domain optical coherence tomography in the 1.3 μm wavelength range," J. Biomed. Opt. **16**, 116020 (2011).
8. C. W. Merkle and V. J. Srinivasan, "Laminar microvascular transit time distribution in the mouse somatosensory cortex revealed by Dynamic Contrast Optical Coherence Tomography," NeuroImage **125**, 350–362 (2016).
9. M. Augustin, S. Fialová, R. Plasenzotti, *et al.*, "Multi-functional OCT image processing for rodent eyes," in *Optical Tomography and Spectroscopy,* (OSA - The Optical Society, Center for Medical Physics and Biomedical Engineering, Medical University of Vienna, Waehringer Guertel 18-20, 4L, Vienna, A-1090, Austria, 2016), p. 3. Export Date: 10 June 2020 Correspondence Address: Augustin, M.; Center for Medical Physics and Biomedical Engineering, Medical University of Vienna, Waehringer Guertel 18-20, 4L, Austria; email: marco.augustin@meduniwien.ac.at.
10. V. J. Srinivasan, J. Y. Jiang, M. A. Yaseen, *et al.*, "Rapid volumetric angiography of cortical microvasculature with optical coherence tomography," Opt. Lett. **35**, 43–45 (2010).
11. D. J. Harper, M. Augustin, A. Lichtenegger, *et al.*, "Retinal analysis of a mouse model of Alzheimer's disease with multicontrast optical coherence tomography," Neurophotonics **7**, 015006 (2020). Export Date: 4 June 2020 Correspondence Address: Harper, D.J.; Medical University of Vienna, Center for Medical Physics and Biomedical EngineeringAustria; email: danielle.harper@meduniwien.ac.at.
12. G. R. Untracht, M. S. Durkee, M. Zhao, *et al.*, "Towards standardising retinal oct angiography image analysis with open-source toolbox octava," Sci. Reports **14** (2024).
13. T. T. Hormel, J. Wang, S. T. Bailey, *et al.*, "Maximum value projection produces better en face oct angiograms than mean value projection," Biomed. Opt. Express **9**, 6412 (2018).
14. D. M. Sampson, A. M. Dubis, F. K. Chen, *et al.*, "Towards standardizing retinal optical coherence tomography angiography: a review," Light. Sci. & Appl. **11** (2022).
15. C. S. Lee, A. J. Tyring, Y. Wu, *et al.*, "Generating retinal flow maps from structural optical coherence tomography with artificial intelligence," Sci. Reports **9** (2019).